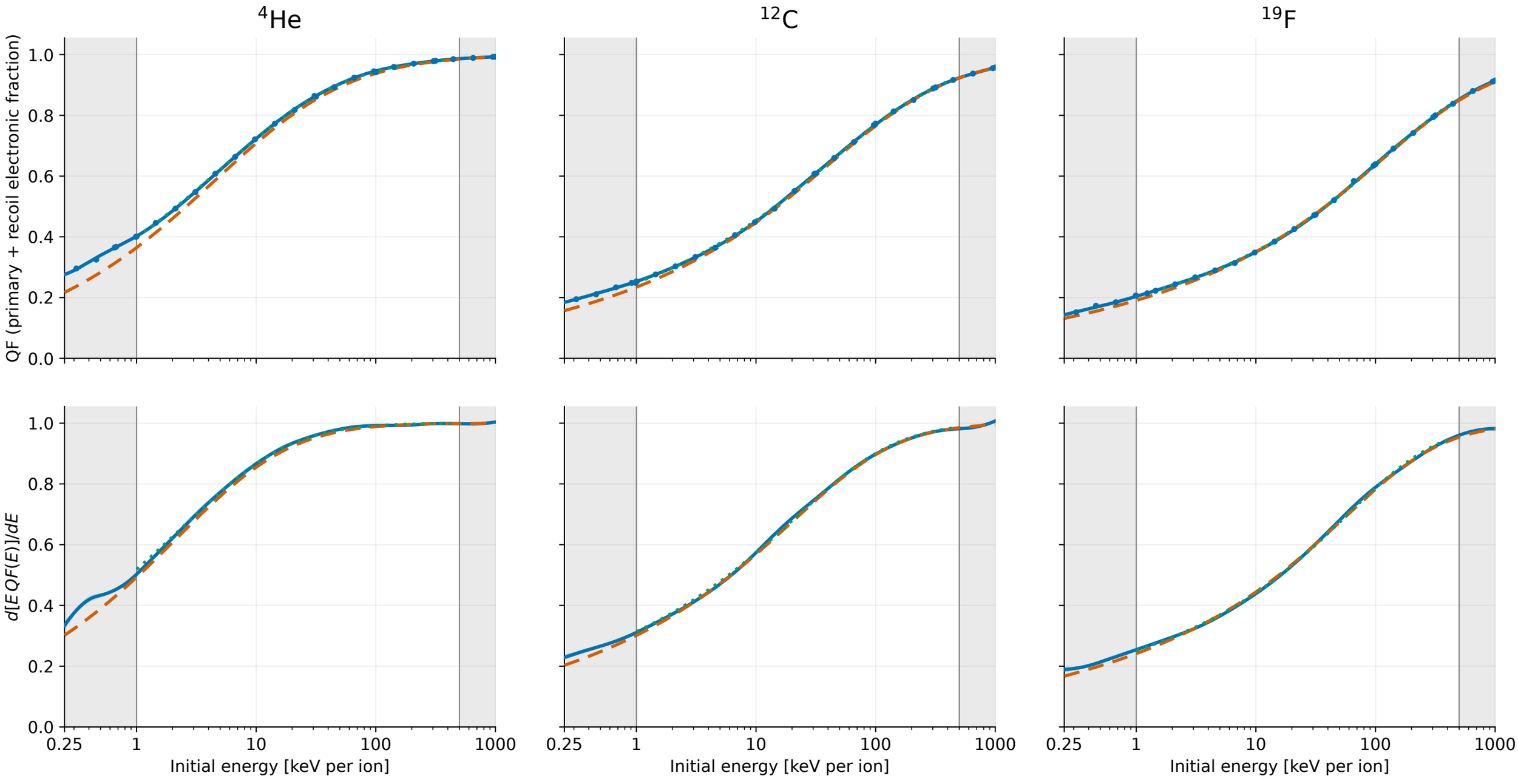
CYGNO quenching recap — thesis implementation vs completed TRIM scan

He/CF_4 60/40 molar · new direct simulations: 900 mbar, 293 K · electronic-energy response

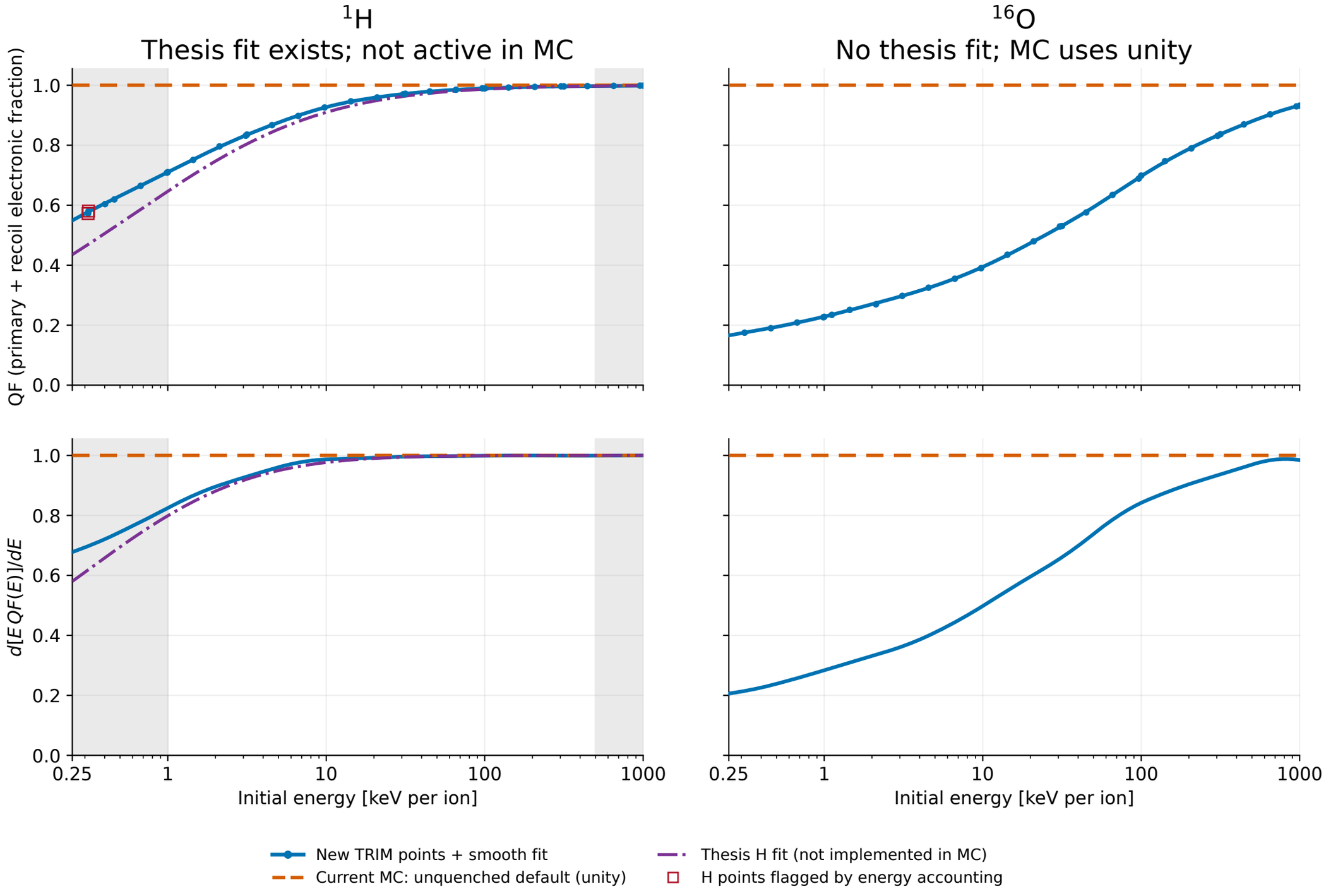


Blue bands: ±1 conditional statistical SEM; systematic/model uncertainty excluded. Green refit is not a validated replacement.

Only He/C/F have fitted QFs implemented in current MC. H/O use unity defaults; the thesis does provide an H fit, but no O fit.

CYGNO recap — new hydrogen and oxygen recoil responses

He/CF_4 60/40 molar · 900 mbar, 293 K · full-cascade TRIM



Blue bands: ±1 conditional statistical SEM; model/calibration uncertainty excluded. H red squares retain ~1.2% closure warnings.

Grey applies to the thesis H fit only: extrapolation outside 1–500 keV. Unity lines are code defaults, not physical QF predictions.

CYGNO QF — thesis cross-check

CYGNO QF recap and thesis cross-check — 16 October 2026

Verdict
The scan reproducibly predicts SRIM electronic energy, including excitation. It is useful as a nominal benchmark, but it is not a calibrated free-charge QF. We cannot claim that no improvement is possible, especially below 1 keV or for the response derivative. No further broad scan is justified by statistics alone.

What the thesis establishes
Di Giambattista's thesis [1], pp. 127–130, sums both primary and secondary-recoil IONIZ bins and divides by initial energy (Eqs. 4.9–4.11): the same definition used here. Table 4.1 exactly matches the He/C/F coefficients in current MC. The thesis simulated 100 ions at 23 energies and fitted 1–500 keV. It explicitly excluded lower energies. Thus the orange curve below 1 and above 500 keV is EXTRAPOLATION. Eq. 4.17, pp. 131–132, defines the same d[E QF(E)]/dE plotted here. Quick mode is stated for preliminary target-width optimization; the thesis does not explicitly identify the production QF mode. It also supplies an H fit that current MC does not use; O is absent. MC H/O unity is therefore a code default.

What was done here
Full mode 5 cascades cover H1/He4/C12/O16/F19 in baseline 60/40 He/CF4 and a second medium with 1000 ppm each H2O/O2. Source conditions: 900 mbar, 293 K, rho=0.001389194959 g/cm3; atomic fractions He=3/13, C=2/13, F=8/13. Requested 293.15 K reuse is approximate. The campaign completed 1614 production/refinement and 180 systematic batches; including pilots/controls: 1939 batches, 180011 histories. All 1939 archives passed integrity checks; independent electronic-energy re-integration differs by less than 0.00845%. Primary/recoil energy was counted once, with separate conservation checks. The final baseline interface has 142 direct points at 0.25–1000 keV. He/C/F each have 12 points at 0.25–10 keV and statistical SEM below 2%. Range/straggling were also calculated; sampled paths are not complete 3D arc lengths. Impurity QF comparisons meet the nominal 2% criterion at 110 eligible points, without a simultaneous-coverage guarantee.

Hydrogen and oxygen in this recap
At 1 keV the direct new QFs are H=0.7102 +/- 0.0011 and O=0.2279 +/- 0.0018. Both new QFs and response derivatives are plotted on the second page. Orange unity lines show current MC defaults; purple is the separate thesis H fit, which is not implemented. No thesis O curve exists. The flagged low-energy H points are marked rather than hidden.

What the discrepancy means
At 1 keV, direct new He/C/F QFs are 0.4011/0.2534/0.2061 versus the old formula's 0.3644/0.2344/0.1908. This 8–10% difference predates smoothing; at 1 keV it cannot be dismissed as extrapolation. Below 1 keV, however, the thesis supplied no fit data. Both methods include secondary electronic losses: missing recoils are not an established explanation. Exact old density/binding inputs remain unavailable. The thesis discusses compound/CAB corrections without a numerical correction; ours uses 1.0. It says default displacement thresholds of tens of eV were retained for topology studies (p. 135), but supplies no element-specific values. Our uniform 21 eV is not proven identical. Changing 21→10 eV shifts 1 keV He/C/F QF by +3.7/+9.4/+11.5%: this shows sensitivity, not the cause of the old/new discrepancy. Low-energy H conservation flags remain. Published old parameter errors lack covariance, so no legacy band has been invented.

Improvement possible without new simulations
The green curves refit the thesis form to new data inside 1–500 keV. Their diagnostic chi2/dof is He 2.96, C 2.90, F 2.58 using new statistical errors only: a useful smoother, but not an exact description. Comparing it with the blue spline exposes derivative model dependence. The derivative wiggles are not directly simulated observables. Retain both as a sensitivity check; do not claim the narrow blue band is total error. Next priorities are matching original inputs and quantifying model/fit uncertainty, then experimental charge calibration. More simulated ions alone cannot settle these.

[1] Flaminia Di Giambattista, Background study of the LIME prototype at underground LNGS for the CYGNO experiment, PhD thesis, 1 March 2024, Sec. 4.1.2.1. Printed pp. 127–135 correspond to PDF pages 136–144; Table 4.1 is PDF page 139. Frozen inputs and hashes, matched-fit details and checks are in plot_evidence.json.